

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code : 5341	Course Title: Big Data Analytics
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To familiarize with Big Data and its sources
- To understand the basics of data preprocessing and data visualization
- To familiarize Big Data tools and techniques.

Course Prerequisites:

Topic	Course code	Course name	Semester
Understand the concept of Database systems	3345	Database Systems Lab	III

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concepts of Big Data	15	Understanding
CO2	Summarize the basic concepts in Big Data Analytics and NoSQL	14	Understanding
CO3	Demonstrate Map Reduce programming in Hadoop Environment	15	Applying
CO4	Make use of basic programming tools like Hive and Pig in Hadoop ecosystems	16	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	Big Data concepts		
M1.01	Summarize the Characteristics of Big Data	2	Understanding
M1.02	Classify Digital Data	2	Understanding
M1.03	Describe Big data Analytics and its applications	2	Understanding
M1.04	Compare Big Data and Data Warehouse	1	Understanding
M1.05	Describe Data Mining Concepts	2	Understanding
M1.06	Discuss Preprocessing techniques	3	Understanding
M1.07	Describe Data Visualization and List some tools used for Visualization	3	Understanding

Contents:

Introduction to Bigdata: Data –Big data –Examples of Big data

Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics.

Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured

Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.

Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods

Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (Basics only)

Data Visualization: Definition, Need for Visualization, List some tools used for Visualization

CO2	Summarize the basic concepts in Big Data Analytics and NoSQL		
M2.01	Explain the importance of Big Data Analytics and the major challenges faced by Big Data.	3	Understanding
M2.02	Outline the concepts of Data Science	2	Understanding
M2.03	Summarize the Big Data Environment terminologies	4	Understanding
M2.04	Summarize the concepts of NoSQL	4	Understanding
M2.05	Outline the concepts of NewSQL	1	Understanding
	Series Test – I	1	

Contents:

Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics

Data Science: Definition , Techniques used in Data Science - classification, regression and clustering

Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems.

NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data.

NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL

CO3	Describe Map Reduce programming in Hadoop Environment		
M3.01	Explain the fundamentals of Hadoop	4	Understanding
M3.02	Concepts of HDFS	1	Understanding
M3.03	List the components of Hadoop Ecosystem	3	Understanding
M3.04	Illustrate the concept of HDFS and MapReduce	2	Applying
M3.04	Use the different function in MapReduce	5	Applying

Contents:

Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions
Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop

Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting.

CO4	Outline the basic concepts of Hive and Pig		
M4.01	Demonstrate the basic concepts of Hive	4	Applying
M4.02	Apply Hive Query Language Statements	3	Applying
M4.03	Make use of Hive User Defined Function	2	Applying
M4.04	Make use of the basic concepts of Pig	4	Applying
M4.05	Make use of HDFS Commands	3	Applying
	Series Test -2	1	
Contents: Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive			

Text / Reference

T/R	Book Title/Author
T1	T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015
T2	Jiawei Han, Champaign Micheline Kamber, Jian Pei Simon Fraser “” Data Mining Concepts and Techniques Third Edition”
R1	Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, Wiley and SAS Business Series, 2012.
R2	Tom White, — Hadoop: The Definitive Guide Third Edition, O’reily Media, 2012
R3	EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012.

Online Resources

Sl.No	Website Link
1	https://hadoop.apache.org/
2	https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/
4	https://hive.apache.org/
5	https://www.tutorialspoint.com/hadoop/index.htm
6	https://pig.apache.org/